Andrew Aburustum

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EDUCATION

Franklin W. Olin College of Engineering — Needham, MA
B.S. in Electrical and Computer Engineering (intended) — *Expected May 2030*

EXPERIENCE

Memorial Sloan Kettering Cancer Center — New York, NY
Biomedical Engineering Intern — *Summer 2025*

- Developed post-processing techniques for biomedical implants and machined parts for medical devices
- Operated Stratasys and resin 3D printers and helped repair a plasma cutter's gantry

Olin College IT Department — Needham, MA
Student IT Technician — *Fall 2026 - Present*

- Work 10-20 help desk tickets a week, swapping components to repair student and staff laptops

ENGINEERING TEAMS

Olin Baja — Needham, MA
Electrical Team Member — *Fall 2026 - Present*

- Co-developing a non-contact fuel-level estimator using accelerometers, a custom PCB, and a Python model

FIRST Tech Challenge Team 14450 RoboRebels — Saddle River, NJ
Team Captain — *2022 - 2026*

- Led the team to win the international 2025 Mexico Premier Event as the 1st seed (6-0)
- Mentored 30 FTC teams across the world, including teams in Australia, Morocco, and Venezuela
- Designed, built, and programmed 3 robots with linear slide, arm, and flywheel auto-targeting systems
- Two-time NJ State Championship finalist; won 2 Innovate and 2 Control Awards

PROJECTS

AI & Machine Learning Projects | *Python, PyTorch, TensorFlow, Keras* — github.com/mumtez/AIfeb14

- Trained VAE, GAN, PixelCNN, and diffusion models in PyTorch to generate weather and Pokémon images
- Developed dictionary-based, statistical, and neural network sentiment classifiers
- Built weather and facial image classifiers

Homelab | *Proxmox, Docker, Portainer, Nginx Proxy Manager, Linux, GRUB/UEFI*

- Build Docker images and Compose stacks with container networks and health checks
- Run services on a Proxmox host through Portainer, routed over HTTPS by Nginx Proxy Manager

FTC Event Viewer | *TypeScript, Docker, FTCScout API* — ftc.aburustum.com

- Built a scouting web app, used by 30+ FTC teams, that ranks event teams by offensive power rating

Console & GPU Repair | *BGA rework, SMD soldering*

- Repaired game consoles and GPUs, replacing HDMI ports and GPU memory chips

SKILLS

Software:	Python, Java, TypeScript, MATLAB, PyTorch, TensorFlow, Keras, Git, Linux, GRUB/UEFI
CAD & PCB Design:	Fusion 360, Onshape, SolidWorks, KiCad, Altium Designer
Fabrication:	3D printing (FDM, resin, Stratasys), machining, SMD soldering, BGA rework, reflow
Languages:	Italian (intermediate, B1), Arabic (elementary, A2)

ACTIVITIES & VOLUNTEERING

Olin Handball, Division 2 Athlete — Fall 2026 - Present
EMS Summer String Festival, Volunteer Violin Camp Counselor — Summers 2023 - Present
Community Chest of Eastern Bergen County, Volunteer — 2018 - Present